

A Market-Clearing Role for Inefficiency on a Limit Order Book

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Abstract

Using a stochastic sequential game, this paper models limit order book trading dynamics. It deduces *ex ante* surplus and some agents' strategies by using an intuitive stationarity property of equilibrium. This largely bypasses any need for direct analysis of agents' (traders') intricate forecasting problems. Surplus per agent, while decreasing in the bid-ask spread, is invariant to some interesting dynamic features of the model. One interpretation of this is that market inefficiency is fixed by the spread at a 'liquidity market-clearing' level. The model best describes cases where price-discreteness leads to a mainly-constant spread. Here, a smaller price tick size raises surplus, and is to be encouraged.

JEL classification: C73, G14, G24

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1 Introduction

Electronic limit order books have become a dominant mode of exchange on liquid financial markets. This paper models them dynamically using game theory. Like the reality, the proposed model has intricate dynamics stemming from fairly simple rules, making it hard to obtain closed-form results about, for example, important questions of market (in)efficiency. While numerical methods have been used elsewhere to handle the intricacy, this paper takes a different tack. Payoffs and actions are simplified, without suppressing the intricacy, so that a subset of equilibrium strategies can be solved for. A way is found here to avoid tackling the complex forecasting problems solved by agents in equilibrium. This yields an expression for each agent's expected payoff *ex ante*, or surplus, on which basis conclusions about market efficiency are drawn.

Above all, the analysis exploits the stationarity of the market microstructure, in particular of offered depths (i.e. of the volume of extant limit orders) at prices near the currently prevailing price. This stationarity implies that although agents (traders) can choose strategically between making and accepting offers to trade, and condition this choice on rich information about the limit order book, the resulting supply of liquidity obeys a simple condition whereby over time there are equally many uncanceled limit orders submitted as market orders. Following *inter alia* Foucault, Kadan, and Kandel (2005), one may think of limit orders as the supply of liquidity (or immediacy), and market orders as the demand; and understand this as a 'liquidity market-clearing' condition.

While surplus per agent is sensitive to the bid-ask spread, it is otherwise invariant to a number of interesting features of the model: including the rate of trader arrival, the quality of traders' information about the limit order book, and the rules for limit order queuing enforced by the exchange. A useful intuition for this is as follows. As a benchmark, consider an equilibrium satisfying the aforementioned market-clearing condition. Suppose that changing the primitives of the model resulted in another equilibrium with greater surplus per agent but the same bid-ask spread. As market orders' pricing and (trivial) execution delay would not thereby change, nor, under the maintained assumptions, would their desirability. Thus the supposed gains in surplus would be enjoyed only by limit order-submitters. So, the market would be oversupplied with agents who prefer to submit limit orders, which in turn would accrue indefinitely, violating the stationarity of offered depths in equilibrium. This is a contradiction. So, the new equilibrium offers no greater surplus. A converse argument shows that it also offers no less.

Surplus per trader in this trading game with private values, may be understood as those gains from trade that are not wasted 1) as limit orders await their counterparties; or 2) in matching

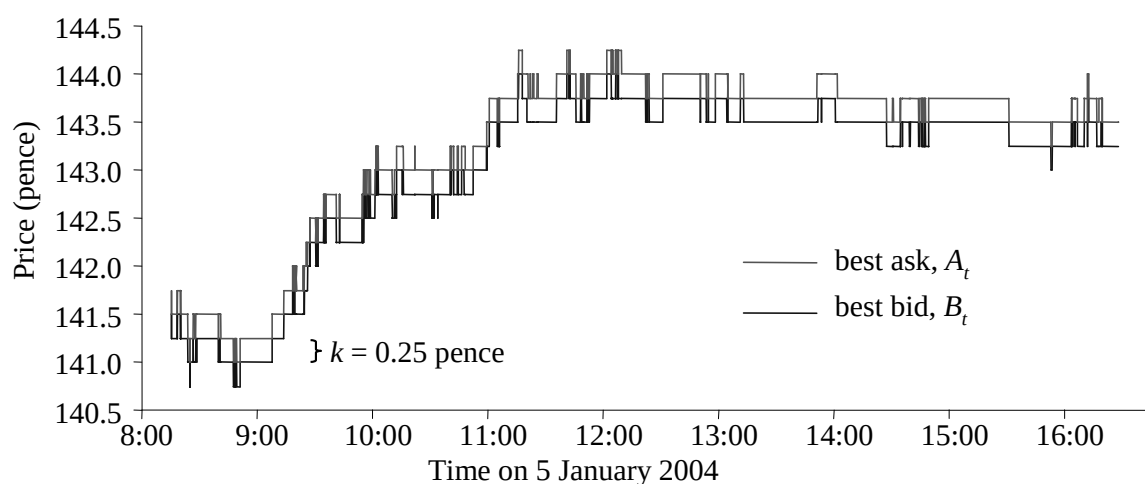


Figure 1: The prices of the best bid and best ask for Vodafone shares on the London Stock Exchange, 5 January 2004. The number of transactions on this day was 14 times higher than the number of changes in the bid (ask). The bid-ask spread was wider than 0.25 pence, only 3.5 per cent of the time. The notation, k , indicates the price tick size.

failures when the asset is bought (sold) by a trader with an excessively low (high) reservation value. The waste is the market's inefficiency, which is therefore identified, via this analysis, by the 'liquidity market-clearing' condition. It is the suggestive idea that inefficiency can but adjust to a market-clearing level which inspired the paper's title. But the broader conclusion is that the stationarity of market microstructure imposes substantive constraints on limit order book dynamics; and aspects of those dynamics may be understood better in its light.

Trader surplus turns out to be invariant to the rate of arrival of traders at the market. The model is at its most realistic when this rate is high. In this setting the asset *trades on a penny*, meaning that the bid-ask spread is bid down to equal its regulatory minimum, the price tick size (a cent, five cents etc.), practically all the time (the bid-ask spread cannot actually be constant: for, it must momentarily widen when the best bid and best ask move up or down). Market depths tend here to be relatively large. This is in fact observed on many important financial markets for interest rates futures, index futures, currency futures, stocks, and other assets.¹ An example from the universe of stocks, is that of Vodafone on the London Stock Exchange, as depicted in Figure 1.

When the asset trades on a penny, the exchange can directly alter the bid-ask spread by adjusting the price tick. The model gives clear normative guidance in this case: the price tick size has a strong effect on market inefficiency and should be cut, at least so far as trading is still

¹Examples of assets whose limit order books traded on a penny in 2006 include: BNP Paribas equity (on Euronext), Vodafone equity (on the London Stock Exchange), US 10-year Treasury Bond Futures (on the Chicago Board of Trade), EURIBOR, Short Sterling and Euro-Swiss Franc Futures (the latter three are on Euronext.liffe).

on a penny. The mechanism via which the tick size impacts welfare works as follows: Being equal to the bid-ask spread, the tick size is the penalty paid for the advantage of immediate execution. Hence, cutting it encourages too many market orders to sustain stationarity, unless limit orders also becomes *ex ante* more attractive – in aggregate, increasing *ex ante* surplus.

In fact, inefficiency per trader is a little under half the price tick size: which is natural, for half of all traders opt to wait for a counterparty by submitting a limit order, and these traders expect the wait to be less costly than paying up the price tick to cross the bid-ask spread, and trading immediately.

The adjustment of inefficiency to market-clearing levels can be understood partly as an adjustment in average depths. The paper analyzes depths' role in this and provides predictions in line with prior empirical findings: that when a limit order book trades on a penny, average inside depths increase with

- trading volume or order intensity (limit order plus market order flow), and
- the price tick size.

Inside depths' positive dependence on trading volume is observed on the Stock Exchange of Hong Kong in Brockman and Chung (1996). See also Danielsson and Payne (2002)² and Lee, Mucklow, and Ready (1993) which, like Kavajecz (1999), focuses on the hybrid New York Stock Exchange. Inside depth's positive dependence on the price tick size on the NYSE was estimated in Harris (1994) and studied in Goldstein and Kavajecz (2000). None of the above studies is of trading on a penny exclusively.

Important theoretical antecedents to this work are in Bernhardt and Hughson (1996), Kadan (2005) and Seppi (1997) which study tick size, inside depths and welfare in dealer and hybrid markets. Foucault, Kadan, and Kandel (2005) explain a channel whereby a large tick size on a limit order book can enhance welfare. A simulation experiment of Goettler *et al.* (2005) suggests that the aggregate effect of a large tick can be to diminish welfare, even when trading is not on a penny. In their model, as here, welfare is decreasing in depth. This runs counter to the intuition that high depths are desirable since unusually large market orders can trade at or near the quotes, which nonetheless is an important consideration in some cases – see Jones and Lipson (2001).

In the prominent works of Kyle (1985) and Glosten and Milgrom (1985), market dynamics are studied when some agents obtain elsewhere superior information about the underlying asset's future cash flows. In the current model, by contrast, some traders do have superior information, but only about future order flow, which they infer rationally from the current market state.

²At a 20 second frequency Danielsson and Payne (2002) gives nuanced evidence on depths and order flow dynamics for Reuters FX markets. It does not fully isolate inside depths.

Therefore there is no winner's curse, or picking-off risk in limit order execution, in the sense developed *inter alia* in Foucault (1999) and Handa and Schwartz (1996).³

The paper proceeds as follows. Section 2 details the model and equilibrium. Section 3 proves the main Theorems about strategies and surplus (detailed derivations are left to an Appendix). Section 4 investigates adaptations to the model, including the introduction of involuntary limit order cancellation. Section 5 derives average inside depths in a simplified model. Section 6 concludes.

2 The Model

The model is a stationary equilibrium in a stochastic sequential game, where as in Foucault *et al.* (2005), Goettler *et al.* (2004) and (2005), Hollifield, Miller, and Sandås (2004), Parlour (1998) and Rosu (2004), impatient traders arrive at a limit order book sequentially and may alter its state. They face a simple trade-off between a market order, and a limit order at a better price which must wait for execution. Other aspects of this trade-off are analyzed in Cohen, Maier, Schwartz, and Whitcomb (1981) and Chakravarty and Holden (1995). The use of stationarity constraints to infer equilibrium best responses in a stochastic sequential game without solving the forecasting or decision problem, is (to the best of my knowledge) a new approach here.

Let there be a probability space called $(\Omega, \mathcal{F}, P)$, with a filtration, $\{\mathcal{F}_t : t \in \mathbb{N}\}$. The duration between successive times t and $(t + 1)$ is the constant $\Delta > 0$.⁴ There is a stochastic process $\chi = \{\chi_t : t \in \mathbb{N}\}$ which is adapted to $\mathcal{F}_t$. In fact, χ_t will play the role of a state variable, and will describe the shape of the limit order book. Now,

$$\chi_t : \Omega \rightarrow \mathbb{Z}^{(Q-M+1)}, \quad (1)$$

where Q and M are natural numbers, and $Q > M$. Let $k \in \mathbb{R}^+$ be the 'price tick'. The limits Qk and Mk are upper and lower inclusive bounds on transaction prices in the model, and they might best be thought of as being quite large and quite far apart. Some preliminary definitions:

³It is expected that moderate asymmetric information about fundamentals *enhances* trading surplus, when the asset trades on a penny. It deters limit orders because of the winner's curse (see Foucault 1999 and Handa and Schwartz 1996). But it leaves the bid-ask spread, and so the expected payoff to a market order, unchanged. This causes a relative reduction in limit order submission, which in turn violates the stationarity property of equilibrium. So, in equilibrium limit orders become more desirable in other ways, enhancing overall surplus – while transferring rents to informational insiders. This may help in thinking about why inside information is scarcely detrimental to welfare in the comparative simulations of similar models in Goettler, Parlour, and Rajan (2004).

⁴In an earlier version of this paper, the duration between successive times t and $(t + 1)$ was an exponentially distributed random variable independent of $\mathcal{F}_t$. This assumption leaves all results of the paper unchanged, except for the existence result in Proposition 3.1. A problem arises in the proof of existence because the assumption implies an uncountable state space.

Definition 1 For all p such that $M < p < Q$, define

$$L_t^{pk} = | \{ \chi_t \}_{(p-M+1)} |. \quad (2)$$

Special care is needed at the endpoints: Define

$$L_t^{Mk} = \{ \chi_t \}_1 \text{ and } L_t^{Qk} = -\{ \chi_t \}_{(Q-M+1)}. \quad (3)$$

Notation 2.1 For any p such that $M \leq p \leq Q$, we will say that at time t the market depth at price pk is L if

$$L_t^{pk} = L. \quad (4)$$

Let $\{ \chi_t \}_{(p-M+1)}$ be the $(p - M + 1)$ 'th element of the vector χ_t . We will say that there are outstanding bids at price pk if

$$\{ \chi_t \}_{(p-M+1)} > 0. \quad (5)$$

If on the other hand

$$\{ \chi_t \}_{(p-M+1)} < 0 \quad (6)$$

we will say that there are outstanding asks at price pk .

Definition 2 Let A_t , a stochastic process giving the **best ask**, be defined by

$$A_t = \inf(\{Qk\} \cup \{pk : \{ \chi_t \}_{(p-M+1)} < 0\}), \quad (7)$$

Intuitively, A_t is the lowest price at which there is an extant ask in the limit order book. Likewise, let B_t , the analogous **best bid**, be defined by

$$B_t = \sup(\{Mk\} \cup \{pk : \{ \chi_t \}_{(p-M+1)} > 0\}). \quad (8)$$

So, we may call $L_t^{A_t}$ the ‘depth at the best ask’, and $L_t^{B_t}$ the ‘depth at the best bid’. Note that A_t and B_t are at all times within the interval $[Mk, Qk]$. These bounds provide a necessary closure to the model, but introduce complexities regarding order book dynamics at the end points, that are dealt with as they arise.

2.1 Players’ actions

One agent (or player, or trader) arrives at each time t . She makes two actions in sequence and then leaves the game forever. Her payoff, which will be defined later, is a random variable that is *not* adapted to $\mathcal{F}_t$ (it depends on future events).

Her first action is to choose a probability π_t from the interval $[0, 1]$. Then a binomial random variable $Z_t \in \{0, 1\}$ is drawn by nature so that with probability π_t , $Z_t = 1$. She observes the product $Z_t\chi_t$. Adopt the convention that if $Z_t = 0$, then $Z_t\chi_t$ is the scalar, 0 (a vector of zeros will be denoted by $\mathbf{0}$, in bold type). So if $Z_t\chi_t = 0$ she may infer that $Z_t = 0$, and that she has no information about the current limit order book, as summarized by χ_t .

Let $\hat{\mathbb{R}}$ be $\mathbb{R} \cup \{-\infty, \infty\}$, so it is the closure of $\mathbb{R}$.

Second, she chooses a cutoff function c_t , conditional on $Z_t\chi_t$.

$$c_t \in \hat{\mathbb{R}}^3, \quad (9)$$

subject to the constraint that c_t is a vector of three weakly increasing elements (which in fact will act as cutoffs determining action).

She has an automatic agent which can submit orders on the limit order book. After she has left the game it compares these cutoffs to her private type (given later). Its intervention thereby depends on them. Proposition 3.2 will show that changing the order of events can yield an observationally identical game in which the notion of an automatic agent would be unnecessary. The reasons for setting up the game this way are to facilitate the proof of equilibrium existence, and to establish the notation for cutoffs, c_t , which will be central to the solution.

Define player t 's overall action to be a_t , so

$$a_t = (\pi_t, c_t). \quad (10)$$

Note that the space of possible choices of c_t is compact.

By way of a preview: I will be interested in an equilibrium where for given χ_t all agents choose a_t identically. The central equilibrium characterization in the paper, is of players' *ex ante* payoff, and of their choice of c_t when $Z_t = 0$.

2.2 Rule for the automatic agent

After the action a_t , player t 's type β_t is drawn randomly from a symmetric distribution on the compact real interval $[-\beta^+, \beta^+]$, independently of all other random variables known at time t . The CDF of this distribution is called F and is a primitive of the model. All other things being equal, $\beta_t < 0$ implies that player t finds selling more desirable than buying, while $\beta_t > 0$ implies the opposite. This will be made explicit in the definition of payoffs.

Player t 's automatic agent then submits an order to the limit order book, denoted or_t , in one of four ways, whose effect on the random variables of the model will be explained shortly. Let

$$or_t \in \{sell, ask, bid, buy\}. \quad (11)$$

Buy and *sell* are market orders, while *bid* and *ask* are limit orders. The automatic agent selects $or_t = or(\beta_t, c_t)$, where

- $or(\beta_t, c_t)$ is *sell* if β_t is less than all three elements of c_t .
- $or(\beta_t, c_t)$ is *ask* if β_t is less than exactly two elements of c_t .
- $or(\beta_t, c_t)$ is *bid* if β_t is less than exactly one elements of c_t .
- $or(\beta_t, c_t)$ is *buy* if β_t is less than none of the elements of c_t .

This makes clear the role of c_t as containing three cutoffs determining action. In the discussion of Proposition 3.2 it will become apparent why or_t is *sell* when β_t is very low, while or_t is *buy* when β_t is very high; and why *ask* and *bid* are at intermediate realizations of β_t . The reasoning for this goes back to the early work of Parlour (1998).

Discussion Agents choose π_t *ex ante*, but they may condition the submitted order on β_t , *ex interim*. How can this be rationalized? In reality, traders cannot acquire exchange-specific information and skills only for the purpose of a single trade, so in the model they commit to π_t without conditioning on their current trading need, i.e. on their type, β_t .

Definition 3 Let $T_t^B \geq 0$ be the random duration after t until at least $(L_t^B + 1)$ sales have been submitted subsequent to t . Let $T_t^A \geq 0$ be the random duration after t until at least $(L_t^A + 1)$ buys have been submitted subsequent to t .

If player t submits a *bid* or an *ask*, then T_t^B and T_t^A are their respective times to execution.

A notational point: throughout, dropping the subscript- t from a random variable will indicate the entire stochastic process to which it belongs. The filtration, $\{\mathcal{F}_t\}$ is the natural filtration of

$$\{(\beta_t, Z_t, \chi_t) : t \in \mathbb{N}\}. \quad (12)$$

However, in proving equilibrium existence we will also be interested in the, more limited, natural filtration of χ and Z . This is because χ and Z (unlike β) are stochastic processes that inhabit countable spaces.

2.3 State transitions

When an order is submitted to the limit order book, this changes the order book shape. A limit order adds to the already-assembled offered depth; while a market order, by trading with an extant limit order, diminishes the depth. So there is a sequence of transitions in the state variable, χ_t , induced by order submission. Furthermore, the state χ_{t+1} is measurable with respect

to χ_t and or_t . It is helpful to think of a function g , such that for all t

$$\chi_{t+1} = g(\chi_t, or_t). \quad (13)$$

Define

- $g(\chi_t, sell)$ is equal to χ_t , except that its depth at B_t is 1 less (so $L_{t+1}^{B_{t+1}} = L_t^{B_t} - 1$).
- $g(\chi_t, buy)$ is equal to χ_t , except that its depth at A_t is 1 less.
- $g(\chi_t, bid)$ is equal to χ_t , except that its depth at price p is 1 greater, where $p = (A_t - k)$ unless $L_t^{B_t} < 0$, when $p = B_t$.
- $g(\chi_t, ask)$ is equal to χ_t , except that its depth at price p is 1 greater, where $p = (B_t + k)$ unless $L_t^{A_t} < 0$, when $p = A_t$.

So, special rules apply in artificial cases of ‘negative depth’, $L_t^{A_t} < 0$ and $L_t^{B_t} < 0$, which arise at the end points when, respectively, $A_t = Qk$ or $B_t = Mk$. Such cases are treated differently to ensure the stationarity of χ .

Finally, the initial condition, χ_0 , is a random variable with a distribution Σ .

Discussion Leaving aside the special rules at the endpoints, this means that *bids* and *asks* can only be submitted at the most aggressive permissible price available. This is a stylized assumption on trader actions which helpfully reduces the number of order types to only four. Some discussion of more discriminating limit pricing is given in Cordella and Foucault (1999) in a model of a limit order book with bid-ask spreads routinely wider than the price tick.

The assumption is most plausible for those assets that trade on a penny, with high offered depths at B_t and A_t . Then, empirically, the most aggressive price possible (as limited by the large price tick size) is chosen by a wide range of trader types.

2.4 Payoffs

Players are risk neutral and discount the future at rate $\rho \in \mathbb{R}^{++}$. Their discounted payoff is $\Pi_t(\beta_t)$ where

$$\Pi_t(b) = \left(b - \frac{k}{2}\right) \mathbf{1}_{or_t=buy} \quad (14)$$

$$+ \left(-b - \frac{k}{2}\right) \mathbf{1}_{or_t=sell} \quad (15)$$

$$+ \left(b + \frac{k}{2}\right) e^{-\rho T_t^B} \mathbf{1}_{or_t=bid} \quad (16)$$

$$+ \left(-b + \frac{k}{2}\right) e^{-\rho T_t^A} \mathbf{1}_{or_t=ask} \quad (17)$$

$$-xZ_t, \quad (18)$$

and $\mathbf{1}$ is the indicator function. So in (18) players incur a penalty $x \in \mathbb{R}^+$, a primitive of the model, for acquiring the advantage that they may condition their order submission on χ_t .

Discussion This payoff may be thought about in the following way: first, consider β_t to be a payoff arising from a private motive to trade, received only when the trade takes place. This may be immediately, in the case of a *buy* or a *sell*; or after a delay, in the case of a *bid* or an *ask*. Before being discounted for such delay, the payoff is *netted of the trade's implementation shortfall, or slippage*. This is in fact realistic: traders' incentives are often set up to penalize any slippage in price, relative to the average of A_t and B_t , known as the mid-quote. While this slippage might vary with fluctuations in the bid-ask spread, $(A_t - B_t)$, this model keeps it simple: slippage is $\frac{k}{2}$ for a *buy* or a *sell*, and $-\frac{k}{2}$ for a *bid* or an *ask*.

In the leading empirical application of this model, the slippage assumption is correct: when markets, such as that depicted in Figure 1, trade on a penny, the bid-ask spread is predominantly equal to k , and hence slippage is $\pm\frac{k}{2}$.

In reality, limit orders typically wait at most a few minutes to execute, not long enough for impatience, as usually understood, to become important. The discount rate, ρ , is therefore more a short-hand for a range of opportunity and capital costs outside the model that the trader undergoes in holding a limit order in the market. One way of introducing ρ is to endow each player with an imminent, unobserved, idiosyncratic liquidity event or deadline.⁵

Agents incur a penalty of x iff $Z_t = 1$. How can this be understood? The penalty can be thought of as the expense, in terms of computing and physical effort, of gathering, organizing, interpreting and acting on timely information about the limit order book. It could also be seen as a fee paid to the exchange for access to current information. However, on this interpretation, the resulting revenues to the exchange must be understood as excluded from the welfare analysis.

2.5 Equilibrium

Traders all have the same strategy, which maps signals, $Z_t\chi_t$, into actions, a_t . The strategy is a pair $(\pi, c(\cdot))$, where $\pi \in [0, 1]$ and c takes as its argument signals of the form $Z_t\chi_t$.

An equilibrium is a probability space $(\Omega, \mathcal{F}, P)$, together with a strategy and the primitives of the model. Equilibrium properties:

⁵On this interpretation, players are patient and do not discount the future. The deadline arrives exogenously as the first event of a Poisson process with parameter ρ . Execution after the deadline results in a zero payoff.

It is possible to formulate an adapted version of the model with the same qualitative properties, where, as in Foucault, Kadan, and Kandel (2005), rather than having a discount rate, traders undergo a constant waiting cost per unit time that their limit order remains in the market before trading.

1. Players choose actions $a_t = (\pi, c(Z_t\chi_t))$. (So $\pi_t = \pi$ and $c_t = c(Z_t\chi_t)$.) The space $(\Omega, \mathcal{F}, P)$ is consistent with the sequence of players' actions $\{a_t : t \in \mathbb{N}\}$.
2. When $a_t = (\pi, c(Z_t, \chi_t))$, player t acts in the way which maximizes the conditional expectation of $\Pi(\beta_t)$, given other players' strategies. Specifically, let $E_{\pi^\dagger, c^\dagger}^t$ indicate the expectations operator when for all $s \neq t$, $a_s = (\pi, c(Z_s\chi_s))$, the equilibrium action; but at time t , $a_t = (\pi^\dagger, c^\dagger)$. Then

$$E_{\pi^\dagger, c(Z_t\chi_t)}^t[\Pi_t(\beta_t)] \quad (19)$$

exists and is maximized when $\pi^\dagger = \pi$. And for all $z \in [\mathbb{Z}^{(Q-M+1)} \cup \{0\}]$

$$E_{\pi, c^\dagger}^t[\Pi_t(\beta_t) | Z_t\chi_t = z] \quad (20)$$

exists and is maximized when $c^\dagger = c(z)$. ■

Lemma 2.2 *Any equilibrium of this game for which $\pi < 1$ has its initial condition drawn from a distribution Σ such that χ is a stationary stochastic process.*

Proof. As $\pi < 1$, sometimes a player, t , receives the signal $Z_t\chi_t = 0$. If χ were not stationary, her decision problem would not then be defined, for she would have no measure over the state space $\mathbb{Z}^{(Q-M+1)}$. ■

Although this stationarity property makes equilibrium existence harder to prove, the benefit is substantial: the stationarity of the state, which will play an important part in the analysis, is guaranteed provided $\pi < 1$. For this reason I concentrate on equilibria where (endogenously) $\pi < 1$, so that not all traders are sophisticated users of order book information.

From now on, an unembellished expectations operator, E , will mean the one when in equilibrium. It will also be concise to suppress the symbol π in $E_{\pi, c^\dagger}^t$, by henceforth writing $E_{c^\dagger}^t = E_{\pi, c^\dagger}^t$.

It helps to define an additional property that the equilibrium of interest will in fact possess:

Definition 4 *For any vector v , let $rev(v)$ be v with its elements reversed. Say a function f is buy-sell symmetric if for any argument, v , $f(-rev(v))$ and $rev(f(v))$ exist, and*

$$f(-rev(v)) = -rev(f(v)). \quad (21)$$

An equilibrium is buy-sell symmetric if c is buy-sell symmetric.

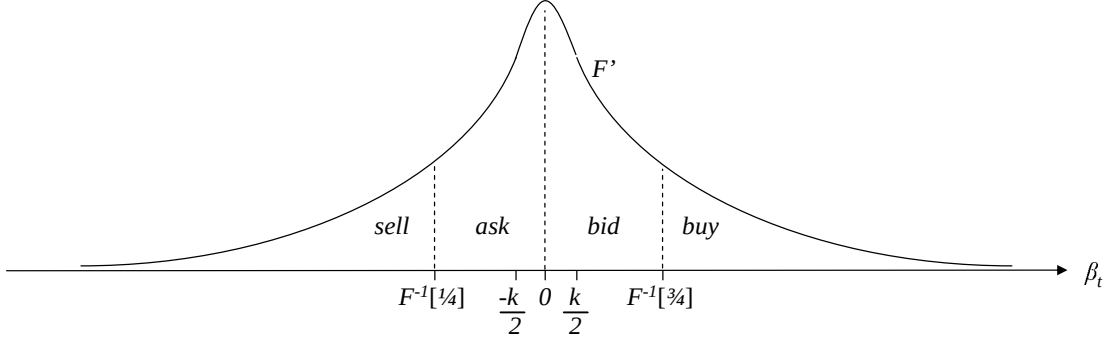


Figure 2: The distribution of the type, β_t , and (as dotted lines) the three equilibrium cutoffs in $c(0)$. The four delimited areas under F' are equal to one another and equal to $\frac{1}{4}$.

Note that $-rev(\chi_t)$ represents a limit order book where the extant bids and asks in the order book at time t have been reversed. Put loosely, Definition 4 states that a buyer responds to χ_t just as a seller would respond to $-rev(\chi_t)$. This is a natural constraint in this symmetric environment where buyers and sellers face very congruent decision problems.

Lemma 2.3 *In a buy-sell symmetric equilibrium where $\pi < 1$, the (ordered) pair of processes (χ, β) has the same distribution as $(-rev(\chi), -\beta)$.*

Proof. See Appendix. ■

2.6 Shape of the CDF, F

The analysis of the game's (as we shall see) intricate equilibria, is only applicable for a specific family of CDFs, F . Suppose that the distribution of β_t is symmetric about zero, and has connected, bounded, support $[-\beta^+, \beta^+]$. Its CDF is F . For all $\beta \in S = (-\beta^+, -\frac{k}{2}]$,

$$F(\beta) = \frac{\eta}{\frac{k}{2} - \beta}. \quad (22)$$

The constant η is chosen to ensure that $F(\infty) = 1$. Furthermore, $F(-\frac{k}{2}) > \frac{1}{3}$. To satisfy (22) there are suitable atoms at $-\beta^+$ and β^+ . This truncation of F simply ensures that moments exist. It should not be too extreme: suppose that

$$\beta^+ > \frac{k}{1 - e^{-\rho\Delta}}. \quad (23)$$

A typical form of F is depicted in Figure 2.

3 Equilibrium characterization

The characterization of equilibrium will proceed in four parts. Section 3.1 proves the existence of equilibrium. Section 3.2 establishes that, when ignorant of the limit order book (i.e. when $Z_t = 0$), traders face a particular *average* problem. In parallel, Section 3.3 deduces from stationarity the average best response. After these preliminary steps, Section 3.4 provides the main theorems of the paper. It shows that the best response, namely $c(0)$, to the average problem just mentioned, is this average best response, which depends in fact on the quartiles of F – as depicted in Figure 2. This identifies $c(0)$ and indirectly thereby surplus per agent.

3.1 Existence of equilibrium

A result of Rieder (1979) can be used to prove the existence of equilibrium.

Proposition 3.1 *For all $\pi \in (0, 1)$, there exists an x , such that a buy-sell symmetric equilibrium exists where $\pi_t = \pi$ for all t .*

Proof. See Appendix. ■

This Proposition's requirement that $\pi > 0$ means that all orders are eventually observed by some traders with positive probability. This ensures sufficient negative feedback in depths for χ to be stationary, which Lemma 2.2 showed was an necessary condition of equilibrium whenever $\pi < 1$. On the other hand, the emphasis on a *buy-sell symmetric* equilibrium is simply an appealing equilibrium selection, although it may be that all equilibria in this game are buy-sell symmetric. The following analysis will concern only buy-sell symmetric equilibria.

Definition 5 *Let c_{ask} be the cutoff function such that the player is sure to submit an ask, so*

$$c_{ask} = (-\infty, \infty, \infty)', \quad (24)$$

and define c_{sell} , c_{bid} , and c_{buy} similarly.

An out-of-equilibrium expectations operator of interest will be $E_{c_{ask}}^t$.

Proposition 3.2 *Consider a buy-sell symmetric equilibrium where $\pi \in (0, 1)$. For any order type, ' $or^\dagger$ ', consider the random variable*

$$E_{c_{or^\dagger}}^t [\Pi_t(\beta_t) | Z_t \chi_t, \beta_t]. \quad (25)$$

For each β_t and $Z_t \chi_t$, its maximum is when $or^\dagger = or(\beta_t, c(Z_t \chi_t))$.

Proof. See Appendix. ■

One can therefore reason as follows: Suppose that player t could observe β_t and Z_t , and on this basis select the order, or_t (*bid*, *ask*, etc.), conditioning her selection on $Z_t\chi_t$. So, she is able to act directly on the exchange, rather than via an automatic agent. She is not thereby constrained by buy-sell symmetry as expressed in Definition 4. Then she would not deviate from the buy-sell symmetric and stationary equilibrium. So, the apparently restrictive structure implied by the automatic agent's implementation of the cutoffs in c_t , is in fact not binding.

3.2 Out-of-equilibrium reasoning when $Z_t = 0$

Notation 3.3 *I will use a superscript-1 to denote the first element of a vector. So I denote the first element of the vector v by v^1 .*

The analysis will be concerned mainly with c_t^1 and $c^1(Z_t\chi_t)$. Proposition 3.2 implies that a player t who knows that $\beta_t = c^1(Z_t\chi_t)$ is at a cutoff where she is indifferent between submitting an *ask* and a *sell*, on observing $Z_t\chi_t$. This can be formalized as follows: from (14) – (18),

$$\left(-c^1(Z_t\chi_t) - \frac{k}{2}\right) = \left(-c^1(Z_t\chi_t) + \frac{k}{2}\right) E_{c_{ask}}^t[e^{-\rho T_t^A} | Z_t\chi_t]. \quad (26)$$

Also note that in equilibrium

$$E_{c_{ask}}^t[e^{-\rho T_t^A} | Z_t\chi_t = 0] = E_{c_{ask}}^t[e^{-\rho T_t^A}], \quad (27)$$

for the fact that $Z_t = 0$, provides no information about the state of the process χ .

Definition 6 *Let the expected discount factor associated with limit order execution be*

$$E_{c_{ask}}^t[e^{-\rho T_t^A}], \quad (28)$$

which, in a buy-sell symmetric equilibrium, is equal to $E_{c_{bid}}^t[e^{-\rho T_t^B}]$.

Proposition 3.4 *In a buy-sell symmetric equilibrium where $\pi \in (0, 1)$*

$$F(c^1(0)) = E[F(c^1(Z_t\chi_t))]. \quad (29)$$

Proof. The upper bound imposed on $-\beta^+$ in (23) ensures that all finite cutoffs $c^1(Z_t\chi_t)$ are identified by (26) and within the support of β_t . Rearranging (26), using the definition of F ,

$$F(c^1(Z_t\chi_t)) = \frac{\eta}{k} \left(1 - E_{c_{ask}}^t[e^{-\rho T_t^A} | Z_t\chi_t]\right). \quad (30)$$

Taking expectations in (30),

$$E_{c_{ask}}^t[F(c^1(Z_t\chi_t))] = \frac{\eta}{k} \left(1 - E_{c_{ask}}^t[e^{-\rho T_t^A}]\right). \quad (31)$$

From (27) and (30), the right-hand side of this is $F(c^1(0))$. So the Theorem follows on observing that since the distribution of $Z_t\chi_t$ is unchanged by a change in c_t ,

$$E[F(c^1(Z_t\chi_t))] = E_{c_{ask}}^t[F(c^1(Z_t\chi_t))]. \quad (32)$$

■

Discussion This proof of Proposition 3.4 follows the reasoning about out-of-equilibrium paths, of a marginal cut-off type who is indifferent between submitting an *ask*, and submitting a *sell*; and who does not observe the state χ_t . It shows that she faces the average of the problems faced by players who observe χ_t , and therefore uses an average cutoff. However this is not an arithmetic average but a deformed average (in fact, it is a hyperbolic average). The proof made no reference to the details of the dynamic equilibrium, in particular to the distribution of T_t^A . The next part shows how stationarity in fact identifies $E[F(c^1(Z_t\chi_t))]$: for, were it higher or lower, depths would be non-stationary.

3.3 The consequences of stationarity

The following Lemma states that in equilibrium *sells*, *buys*, *bids* and *asks* are equally prevalent to one another in the order flow. This formalizes one of the paper's main intuitions: namely, that if depths are stationary then the average number of limit orders per unit time equals the average number of market orders. As discussed in the Introduction, this can be interpreted as a 'liquidity market-clearing' condition.

Definition 7 Let $D_t = \sum_{p=M}^Q L_t^{pk}$. So, D_t is the total market depth at time t .

Lemma 3.5 In a buy-sell symmetric equilibrium where $\pi \in (0, 1)$, for any t

$$\Pr[or_t = sell] = \Pr[or_t = ask] = \Pr[or_t = bid] = \Pr[or_t = buy] = \frac{1}{4}. \quad (33)$$

Proof. Since χ is stationary, D is also. D_{t+1} always differs from D_t by exactly 1. So, D rises as often as it falls. As it falls exactly when $or_t \in \{sell, buy\}$, for all t

$$\Pr[or_t \in \{sell, buy\}] = \frac{1}{2}. \quad (34)$$

In a buy-sell symmetric equilibrium, χ and $-rev(\chi)$ have the same distribution. So, *buy* and *sell* are equally prevalent orders. And, *bid* and *ask* are equally prevalent orders. ■

Since *sells*, or market sales, are submitted by those players who draw β_t below $c^1(Z_t\chi_t)$,

$$\Pr[or_t = sell|Z_t\chi_t] = F(c^1(Z_t\chi_t)). \quad (35)$$

Given Lemma 3.5 this implies (by the Law of Iterated Expectations) that

$$E[F(c^1(Z_t\chi_t))] = \frac{1}{4}. \quad (36)$$

Think of $F^{-1}(\frac{1}{4})$ as the median seller. Then (36) shows that because of stationarity, in particular of the stationarity of offered depths, the median seller is (just like $c^1(0)$ although for quite different reasons), a deformed average of the set of marginal types, $\{c^1(Z_t\chi_t)\}$ (it might have been a simple arithmetic average if F had been a uniform CDF). In the light of this, it is perhaps a natural thought that the median seller would be roughly indifferent between a *sell* and an *ask* in the average of their problems, as faced by traders with $Z_t = 0$. In fact, features of F imply that this deformation is the same as the one in Proposition 3.4, so that this natural thought is exactly true. This is formalized in Theorem 3.6.

3.4 The main theorems

Theorem 3.6 *In a buy-sell symmetric equilibrium where $\pi \in (0, 1)$,*

$$c(0) = \begin{pmatrix} F^{-1}(\frac{1}{4}) \\ 0 \\ F^{-1}(\frac{3}{4}) \end{pmatrix}. \quad (37)$$

Proof. From Proposition 3.4 and (36), $c^1(0) = F^{-1}(\frac{1}{4})$. The theorem then follows from the buy-sell symmetry of $c(0)$. ■

So it has been possible to identify the best response to an absence of limit order book information ($Z_t = 0$), while bypassing the out-of-equilibrium reasoning for that best response almost entirely. Such reasoning is intricate and involved, for it reviews each possible market state; and for that state, it considers the market's likely subsequent evolution. Tackling it head-on might be infeasible. The derived best response is illustrated in Figure 2.

Theorem 3.6 shows that the types who have $\beta_t = \pm F^{-1}[\frac{1}{4}]$, are, when $Z_t = 0$, indifferent between 1) a limit order, and 2) a market order which incurs k in extra slippage. This actually identifies the *expected discount factor associated with limit order execution*, namely $E_{c_{ask}}^t[e^{-\rho T_t^A}]$, for, from (26),

$$\left(-F^{-1}\left(\frac{1}{4}\right) - \frac{k}{2}\right) = \left(-F^{-1}\left(\frac{1}{4}\right) + \frac{k}{2}\right) E_{c_{ask}}^t[e^{-\rho T_t^A}], \quad (38)$$

so that

$$E_{cask}^t[e^{-\rho T_t^A}] = 1 - \frac{k}{\frac{k}{2} - F^{-1}\left[\frac{1}{4}\right]}. \quad (39)$$

Furthermore, it leads to a characterization of each agent's expected payoff:

Definition 8 *Let*

$$\delta = \frac{E[|\beta_t| : |\beta_t| < F^{-1}\left[\frac{3}{4}\right]]}{F^{-1}\left[\frac{3}{4}\right]}, \quad (40)$$

so that δ is a measure of F 's relative dispersion within its interquartile range, $[-F^{-1}\left[\frac{3}{4}\right], F^{-1}\left[\frac{3}{4}\right]]$.

Theorem 3.7 *In a buy-sell symmetric equilibrium where $\pi \in (0, 1)$ the ex ante expected payoff, or surplus, of player t is given by*

$$E[\Pi(\beta_t)] = E(|\beta_t|) - \frac{k}{2} \left(\frac{\frac{k}{2} + \delta F^{-1}\left[\frac{3}{4}\right]}{\frac{k}{2} + F^{-1}\left[\frac{3}{4}\right]} \right), \quad (41)$$

Proof. See Appendix for the full proof. From Theorem 3.6, any player t with $Z_t = 0$ foresees *ex ante*, that $or_t \in \{bid, ask\}$ iff β_t is in the interquartile range of F , and so

$$\Pr\{or_t \in \{bid, ask\} | Z_t = 0\} = \frac{1}{2}. \quad (42)$$

Furthermore, the cases where β_t is in the interquartile range of F are exactly the cases where her payoff to trading is discounted due to delay. But, the expected discount factor for these cases is known from (39). Combining these observations, her *ex ante* surplus follows after algebra.

Were it not for the cost x , this would be exceeded by the surplus of those for whom $Z_t = 1$; but, since players mix between $Z_t = 0$ and $Z_t = 1$, in fact x absorbs exactly this excess. *Ex ante* surplus is therefore equal to expected surplus conditional on $Z_t = 0$. ■

Interpretation and comparative statics Theorem 3.7 shows that surplus per agent in this market is strictly less than $E[|\beta_t|]$. This bound, $E[|\beta_t|]$, has the following interpretation. Consider an ideal market where no trader waits to trade, and each trade has the property that the buyer, say player t , had $\beta_t \geq 0$ while the seller, say s , had $\beta_s \leq 0$. Then the trading surplus achieved with each trade would be

$$E[(\beta_t - \beta_s) | \beta_t \geq 0; \beta_s \leq 0], \quad (43)$$

or $2E[|\beta_t|]$. Per trader, the surplus would be $E[|\beta_t|]$.

Both of these ideal conditions are violated by the electronic limit order book in practice. Theorem 3.7 quantifies the resulting shortfall in surplus, or inefficiency per trader; and shows the shortfall to be a necessary condition for stationarity. It is

$$\frac{k}{2} \left(\frac{\frac{k}{2} + \delta F^{-1}\left[\frac{3}{4}\right]}{\frac{k}{2} + F^{-1}\left[\frac{3}{4}\right]} \right). \quad (44)$$

Given F and k , (so that the bid-ask spread is held fixed) this shortfall or inefficiency is invariant to changes in the other primitives of the model which may freely alter either

- (*Order Flow*) the duration between trader arrivals, Δ , or
- (*Quality of Trader Information*) the endogenous proportion of traders, π , who observe χ_t .

3.5 Benefits to cutting the price tick size

Theorem 3.7 shows that the inefficiency per trader lies between $\delta \frac{k}{2}$ and $\frac{k}{2}$. Holding fixed $F^{-1} \left[\frac{3}{4} \right]$ and δ , it is increasing in the price tick, k (given the relationship of F to k in (22) it is not strictly possible to hold F fixed in this thought experiment where k changes).

In (44), the half-spread bounds the inefficiency per trader above. An explanation for this is that exactly half of those traders drawing $Z_t = 0$ wait inefficiently before trading; and when a trader does wait, she finds paying the spread *ex ante* weakly costlier than waiting.

Cutting the tick size affects surplus per player because it makes market orders strictly more attractive. All other things remaining equal, some types would be induced to submit market orders, who previously preferred limit orders. This would violate Theorem 3.6 and imply a shortage of limit orders over time, non-stationary depths and no equilibrium. In equilibrium, therefore, endogenous frictions adjust to restore the types in $c(0)$ to their respective indifference conditions as illustrated in Figure 2 (for example, average depths fall). Hence, perhaps counter-intuitively, like market orders limit orders also become *ex ante* more attractive when the bid-ask spread falls – which necessarily involves an increase in *ex ante* welfare per player.

4 Comments on the model and alternative formulations

This Section provides some comments about adaptations to the model which weaken or change leading assumptions. It addresses the effect of cancellation on the model, the consequences of a change in the queuing rules enforced by the exchange, and the robustness of the model to the functional form of F .

4.1 The functional form of F

Proposition 3.4 depends importantly on the specification of the functional form of F in Section 2.6. On this Proposition hangs the main Theorems 3.6 and 3.7. It is therefore important to understand the sensitivity of the model to this tractability assumption. This Subsection address the issue in two ways. First, it consider the case where the CDF of β_t is a uniform distribution, F_u , and consider the modification this brings to the results. Second, it shows that given any

symmetric distribution for β_t with support $[-\beta^+, \beta^+]$, Theorems 3.6 and 3.7 are true in the limit of a sequence of equilibria such that $\pi \rightarrow 0$.

4.1.1 A uniform distribution for β_t

Consider a buy-sell symmetric equilibrium where $\pi \in (0, 1)$, and where β_t are i.i.d. random variables distributed uniformly between $-\beta^+$ and β^+ . Call the CDF of β_t , F_u . Let F , not now a primitive of the model, be a function satisfying the conditions in Section 2.6. Then, tracing through its proof, it can be seen that Proposition 3.4 is still true: $F(c^1(0)) = E[F(c^1(Z_t\chi_t))]$. Equally, Lemma 3.5 is also still true: the four order types are equally prevalent in the order flow. However, it follows from Lemma 3.5 now that, in contrast to (36), $E[F_u(c^1(Z_t\chi_t))] = \frac{1}{4}$. Rearranging,

$$c^1(0) - F_u^{-1}\left(\frac{1}{4}\right) = F^{-1}\left(E[F(c^1(Z_t\chi_t))]\right) - F_u^{-1}\left(E[F_u(c^1(Z_t\chi_t))]\right), \quad (45)$$

So $c^1(0)$ differs from the median seller, $F_u^{-1}\left(\frac{1}{4}\right)$, by the quantity on the right-hand-side of (45). This has the following interpretation: $F_u^{-1}\left(E[F_u(c^1(Z_t\chi_t))]\right)$ is the arithmetic mean of the finite elements of $\{c^1(Z_t\chi_t)\}$. On the other hand, $F^{-1}\left(E[F(c^1(Z_t\chi_t))]\right)$ is a distorted, hyperbolic, mean of them. Therefore, insofar as the hyperbolic mean approximates the arithmetic mean, Theorem 3.6 is approximately true.

4.1.2 A limit as $\pi \rightarrow 0$

Suppose now that β_t are i.i.d. random variables, distributed symmetrically about zero with support $[-\beta^+, \beta^+]$. Call the CDF of β_t simply F . Consider a sequence of buy-sell symmetric equilibria with $\pi \in (0, 1)$ of a sequence of games where in the limit $\pi \rightarrow 0$, so that as the sequence progresses fewer and fewer traders observe χ_t . This is somewhat related to the idea of an exchange ‘going dark’, a phenomenon which has occurred, see Hendershott and Jones (2005).

Note that in the limit,

$$F(c^1(0)) - \Pr[or_t = sell] \rightarrow 0, \quad (46)$$

for in a market where almost all traders observe $Z_t\chi_t = 0$, the events $[or_t = sell]$ and $[\beta_t \leq c^1(0)]$ are almost the same. But Lemma 3.5 is true of every equilibrium in the sequence. Hence

$$c^1(0) \rightarrow F^{-1}\left(\frac{1}{4}\right). \quad (47)$$

Therefore, in the limit Theorem 3.6 is true. So, in the limit $F^{-1}\left(\frac{1}{4}\right)$ is indifferent between $[or_t = ask]$ and $[or_t = sell]$. Therefore, in the limit (39) is true. Finally, the proof of Theorem 3.7 is true as stated, for the limit as $\pi \rightarrow 0$.

4.2 Other queuing rules for limit orders

In the model, as almost universally in reality, limit orders execute in sequence according to price priority. When, however, they are equally priced, price priority must be supplemented by a further criterion: Definition 3 implies that they then execute in the order in which they were submitted.

Other queuing rules than this time-priority (or FIFO) rule can be envisaged, and are sometimes used. For example, some markets operate on a *pro rata* system, whereby *all* limit orders at the same price trade simultaneously against each countervailing market order. The market order is normally too small to trade with them all in their entirety. So, the trade is shared among the limit orders in proportion to their size.⁶ The Euronext.liffe markets for futures in Euro-Swiss Franc foreign exchange, in EURIBOR, and in Short Sterling are all markets that in 2006 traded on a penny using a *pro rata* algorithm.

This can be approximately implemented within the current model by replacing Definition 3. Suppose, rather, that an independent, fair coin-flip is used to allocate each *sell* (*buy*) to the extant bids at B_t (asks at A_t). As in the main model, let T_t^A and T_t^B be the random times to execution of a *bid* and an *ask* at time t , respectively. How would this change the results of the model?

Since T_t^A and T_t^B are not now measurable with respect to $\mathcal{F}^{x,Z}$ the proof of equilibrium existence does not follow as stated. However, existence aside, the characterizations of equilibrium in Theorems 3.6 and 3.7 are true. Tracing through the proofs of these Theorems, it can be seen that no distributional properties of T_t^A and T_t^B are invoked. So the proofs remain valid under the altered distribution.

We may conclude, subject to a concern about equilibrium existence, that trader surplus is unchanged by the altered queuing rules. An intuition for this was given in the Introduction.

4.3 Cancellation

This Section discusses an adaptation to the model where periodically limit orders are cancelled involuntarily. This modelling feature is present in Goettler *et al.* (2005) and Hollifield *et al.* (2004), but is replaced with an endogenous cancellation decision in the simulations in Goettler *et al.* (2004). Deliberate, endogenous, cancellation of limit orders, while possibly the more realistic case, is harder to analyze in this framework, and would be an interesting exercise for future work.

Adapt the model so that there exists $\gamma \in [0, 1)$, such that at each time t , each extant bid and

⁶The exchange uses rounding rules and a coin-flip to distribute ‘odd lots’ arising due to quantity discreteness.

ask is independently removed from the limit order book with probability γ , without payoff to its submitter. The time to execution of the remaining limit orders is appropriately reduced, as they advance in the queue to trade.

Although such involuntary cancellation may seem to be detrimental to traders, in fact it is advantageous for trader surplus. I discuss the intuition for this only. When $\gamma = 0$ we revert to the original game. Consider a buy-sell symmetric equilibrium where $\gamma > 0$. As some limit orders are now removed before they trade, stationarity implies that initially limit orders are voluntarily submitted *more* frequently than market orders, and therefore more frequently than in equilibrium when $\gamma = 0$. Since setting $\gamma > 0$ does not change the payoff to a market order, if more limit orders are to be submitted, they must become more attractive to traders in this equilibrium when $\gamma > 0$, than when $\gamma = 0$. This implies that there is a higher surplus per agent.

So, the dominant effect of involuntary cancellation is the positive externality for *other* limit orders when a limit order is removed, causing them to trade sooner.⁷

5 Average depths

As mentioned in the Introduction, surplus per agent may be thought of as the gains from trade that are not lost 1) in the wait as limit orders await their counterparties; or 2) in matching failures when the asset is bought (sold) by a trader with an excessively low (high) reservation value. The first of these, waiting, is closely correlated with market depths. Therefore the identification of market inefficiency in Theorem 3.7 may also be informative about average depths.

This Section studies the stationary distribution of depths in a special case to understand better their sensitivity to parameters in the trading environment. It concentrates simply on their first moment, namely average inside depth: and shows that (to a first order approximation) this does indeed adjust in ways consistent with the invariances implied by Theorems 3.6 and 3.7. The Section is in this way complementary to the study in Parlour (1998) of depths' transitions from one state to the next. The special case of the model that will be analyzed has these features:

- The market is fairly efficient, so that $e^{-\rho T_i^A} \approx 1$.
- The market is a 'two-tick' market, where $Q = (M+1)$, so that there are only two admissible prices for trade: Mk and Qk .
- In equilibrium $\pi \approx 0$.⁸ Theorem 3.6 shows that when $Z_t = 0$, player t sells iff $\beta_t < 0$.

⁷A full proof of this (which would unfortunately require an involved reformulation of the model) proceeds via the intermediate result that if $\gamma > 0$ then $c^1(0) < F^{-1}(\frac{1}{4})$.

⁸More precisely, as in Section 4 one may think of a sequence of equilibria such that $\pi \rightarrow 0$)

Therefore, when $\pi \approx 0$ the inefficiencies arising due to badly-matched counterparties are almost zero.

If the market is not two-tick, then limit orders risk becoming ‘stale’ if A_t and B_t move far away before they execute. Their submitters may be compensated for this by lower inside depths. Where $\pi \gg 0$, so that a substantial number of traders obtain order book information, average depths decline where such traders avoid adding to deep order books, but rise where traders are drawn to supply liquidity to order books of low depths. A characterization of average inside depth under these generalizations is beyond the current scope.

Definition 9 Define $L_t^{A_t+}$ for $\max\{L_t^{A_t} + 1, 0\}$. So $L_t^{A_t+}$ is a measure of depth, which abstracts from the occasional negative depths permitted at endpoints. Define $L_t^{B_t+}$ likewise.

Recall (39), which gave the equilibrium expected discount factor associated with a limit order:

$$1 - E_{c_{ask}}^t[e^{-\rho T_t^A}] = \frac{k}{\frac{k}{2} - F^{-1}\left[\frac{1}{4}\right]}.$$

Via a Taylor expansion, we have that

$$1 - e^{-\rho T_t^A} = \rho T_t^A + O_p((\rho T_t^A)^2). \quad (48)$$

As $e^{-\rho T_t^A} \approx 1$, $(\rho T_t^A)^2$ is small, and

$$1 - E_{c_{ask}}^t[e^{-\rho T_t^A}] \approx \rho E_{c_{ask}}^t[T_t^A]. \quad (49)$$

$E_{c_{ask}}^t[T_t^A]$ is the expected time until $L_t^{A_t+}$ future *buys* have been submitted. The duration between the consecutive arrivals of traders is Δ . Consider a sequence of equilibria where $\pi \rightarrow 0$. In the limit, the probability of each future trader submitting a *buy* is $\frac{1}{4}$. Hence in the limit

$$E_{c_{ask}}^t[T_t^A] = 4\Delta E[L_t^{A_t+}]. \quad (50)$$

Combining (39), (48) and (50), when $\pi \approx 0$

$$E[L_t^{A_t+}] \approx \frac{1}{4\Delta\rho} \left(\frac{k}{\frac{k}{2} - F^{-1}\left[\frac{1}{4}\right]} \right). \quad (51)$$

Although this analysis has considered the waiting time of an *ask* at A_t , an equivalent analysis and the same approximation hold when A_t is replaced with B_t .

Comparative Statics On the basis of (51) some comparative statics can be derived for the equilibrium average (positive) depths at the best ask, as captured by $E[L_t^{A+}]$, which might also be called the average inside depths:

- Average inside depth is increasing in the tick size, k . This is because, as discussed in the Introduction, a greater tick size would cause a shortage of market orders, unless depths rise to make limit orders less attractive. This effect was observed in Goldstein and Kavajecz (2000) and Harris (1994).
- Average inside depth is increasing in the order flow, $\frac{1}{\Delta}$. An increased trader arrival rate, or order flow, would clear a given depth faster. All other things being equal, this would attract too many limit orders to sustain stationarity. To offset this, in equilibrium depths rise. This corresponds to a finding in Brockman and Chung (1996) for the Stock Exchange of Hong Kong.
- But average inside depth is decreasing in trader impatience, ρ , since impatient traders avoid limit orders, unless induced not to by low depths.

6 Conclusion

The market-clearing condition is a central means of analyzing prices without modelling market microstructure details directly. In focusing on these details, this paper rediscovers market-clearing in a different context: in a minimal stationarity property of microstructure dynamics, meaning that a limit order book induces in expectation equal numbers of market orders and uncanceled limit orders over time. This stationarity is harnessed to infer the best response and so the surplus of traders who do not observe the market state, while bypassing altogether their difficult forecasting problem. Applying this approach to stochastic sequential games with binary action, such as queueing, may offer an intriguing opening for future research.

Equilibrium effects lead to striking invariances in trader surplus: with respect to changes in the sophistication of trading (π); in the rate of order flow (Δ); and in limit order queueing rules. In every such case, the market's endogenous inefficiency does not vary, for its anticipated sensitivity to the change is perfectly offset by endogenous equilibrium adjustments, for example in average depths.

But trader surplus is decreasing in the price tick. Cutting the tick improves the prospects of a market order, thereby overcoming traders' incentives to add their limit orders to an already-congested limit order book. Trader surplus is thereby enhanced.

This provides an intuition for using the quoted or effective spread as a measure of market quality: for it shows that a wide bid-ask spread not only penalizes market orders but also penalizes

limit orders, via equilibrium effects (although it might, on the face of it, appear to favor them). Extending this insight to other forms of market microstructure institution, which might involve introducing a market-clearing role for bid-ask spreads as well as for market inefficiency and depths, represents another future research avenue of considerable interest.

References

- Bernhardt, D. and E. Hughson (1996). Discrete pricing and the design of dealership markets. *Journal of Economic Theory* 71, 148–182.
- Brockman, P. and D. Chung (1996). An analysis of depth behavior in an electronic order-driven environment. *Journal of Banking and Finance* 51, 1835–1861.
- Chakravarty, S. and K. Holden (1995). An integrated model of market and limit orders. *Journal of Financial Intermediation* 4, 213–241.
- Cohen, K., S. Maier, R. Schwartz, and D. Whitcomb (1981). Transaction costs, order placement strategy, and existence of the bid-ask spread. *Journal of Political Economy* 89, 287–305.
- Cordella, T. and T. Foucault (1999). Minimum price variation, time priority and quote dynamics. *Journal of Financial Intermediation* 8, 141–173.
- Danielsson, J. and R. Payne (2002). Liquidity determination in an order driven market. Working Paper, London School of Economics.
- Foucault, T. (1999). Order flow composition and trading costs in a dynamic limit order market. *Journal of Financial Markets* 2, 99–134.
- Foucault, T., O. Kadan, and E. Kandel (2005). Limit order book as a market for liquidity. *Review of Financial Studies* 18, 1171–1217.
- Glosten, L. R. and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.
- Goettler, R., C. Parlour, and U. Rajan (2004). Information acquisition in a limit order market. Working paper Carnegie Mellon University.
- Goettler, R., C. Parlour, and U. Rajan (2005). Equilibrium in a a dynamic limit order market. *Journal of Finance* 60, 2149–2192.
- Goldstein, M. and K. Kavajecz (2000). Eighths, sixteenths, and market depth: changes in tick size and liquidity provision on the NYSE. *Journal of Financial Economics* 56, 125–149.
- Handa, P. and R. Schwartz (1996). Limit order trading. *Journal of Finance* 51, 1835–1861.

- Harris, L. (1994). Minimum price variations, discrete bid-ask spreads, and quotation sizes. *Review of Financial Studies* 7, 149–178.
- Hendershott, T. and C. Jones (2005). Island goes dark: Transparency, fragmentation, and regulation. *Review of Financial Studies* 18, 743–793.
- Hollifield, B., R. A. Miller, and P. Sandås (2004). Empirical analysis of limit order markets. *Review of Economic Studies* 71, 1027–1063.
- Jones, C. and M. Lipson (2001). Sixteenths: Direct evidence on institutional execution costs. *Journal of Financial Economics* 58, 253–278.
- Kadan, O. (2005). So who gains from a small tick size? *Journal of Financial Intermediation* 15, 32–66.
- Kavajecz, K. (1999). A specialist’s quoted depth and the limit order book. *Journal of Finance* 54, 747–771.
- Kyle, A. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1336.
- Lee, C., B. Mucklow, and M. Ready (1993). Spreads, depths and the impact of earnings information: an intraday analysis. *Review of Financial Studies* 6, 345–374.
- Parlour, C. (1998). Price dynamics in limit order markets. *Review of Financial Studies* 11, 789–816.
- Rieder, U. (1979). *Equilibrium Plans in Non-Zero-Sum Markov Games*. In O. Moeschlin and D. Pallaschke (Eds.), *Game Theory and Related Topics*, pp 91-101. North Holland Publishers.
- Rosu, I. (2004). A dynamic model of the limit order book. PhD Thesis, MIT.
- Seppi, D. (1997). Liquidity provision with limit orders and strategic specialists. *Review of Financial Studies* 10, 103–150.

A Appendix

A.1 Proof of Lemma 2.3

The lemma states: In a buy-sell symmetric equilibrium where $\pi < 1$, the (ordered) pair of processes (χ, β) has the same distribution as $(-rev(\chi), -\beta)$.

Proof. Note that

$$\chi_{t+1} = g(\chi_t, or(\beta_t, c(Z_t \chi_t))) \quad \text{and} \quad \beta_{t+1} \perp\!\!\!\perp \chi_{t+1}. \quad (52)$$

If c is buy-sell symmetric, this implies that

$$-rev(\chi_{t+1}) = g(-rev(\chi_t), or(-\beta_t, c(-rev(Z_t\chi_t)))) \quad \text{and} \quad -\beta_{t+1} \perp\!\!\!\perp -rev(\chi_{t+1}). \quad (53)$$

So the conditional distributions

$$(\chi_{t+1}, \beta_{t+1}) | (\chi_t, \beta_t) \quad (54)$$

and

$$(-rev(\chi_{t+1}), -\beta_{t+1}) | (-rev(\chi_t), -\beta_t) \quad (55)$$

are the same. Thinking of these as transition densities of a Markov process, it follows that the stationary distributions of (χ_t, β_t) and $(-rev(\chi_t), -\beta_t)$ are the same. These stationary distributions are the distributions of the initial conditions (χ_0, β_0) and $(-rev(\chi_0), -\beta_0)$, respectively.

■

A.2 Proof of Proposition 3.2

The proposition states: Consider a buy-sell symmetric equilibrium where $\pi \in (0, 1)$. For any order type $or^\dagger$, consider the random variable

$$E_{c_{or^\dagger}}^t[\Pi_t(\beta_t) | Z_t\chi_t, \beta_t]. \quad (56)$$

For each β_t and $Z_t\chi_t$, its maximum is when $or^\dagger = or(\beta_t, c(Z_t\chi_t))$.

Proof. Looking at (14) – (18), (56) takes the values:

$$\left(\beta_t - \frac{k}{2}\right) - xZ_t \quad \text{if} \quad or^\dagger = buy \quad (57)$$

$$\left(-\beta_t - \frac{k}{2}\right) - xZ_t \quad \text{if} \quad or^\dagger = sell \quad (58)$$

$$\left(\beta_t + \frac{k}{2}\right) E_{c_{or^\dagger}}^t[e^{-\rho T_t^B} | Z_t\chi_t] - xZ_t \quad \text{if} \quad or^\dagger = bid \quad (59)$$

$$\left(-\beta_t + \frac{k}{2}\right) E_{c_{or^\dagger}}^t[e^{-\rho T_t^A} | Z_t\chi_t] - xZ_t \quad \text{if} \quad or^\dagger = ask. \quad (60)$$

It is straightforward from the definition of the function $or(b, c)$ in Section 2.2 that there exists a vector, $\tilde{c}(Z_t\chi_t)$, of three weakly increasing elements, such that (56) is maximized when $or^\dagger = or(\beta_t, \tilde{c}(Z_t\chi_t))$. So, at this maximum (56) is equal to

$$E_{\tilde{c}(Z_t\chi_t)}^t[\Pi_t(\beta_t) | Z_t\chi_t, \beta_t], \quad (61)$$

for all β_t and $Z_t\chi_t$. It remains to show that $\tilde{c} = c$.

In equilibrium, for all $Z_t\chi_t$, the expectation

$$E_{c^\dagger}^t[\Pi_t(\beta_t)|Z_t\chi_t] \quad (62)$$

is maximized when $c^\dagger = c(\beta_t, Z_t\chi_t)$. By the Law of Iterated Expectations it is

$$E_{c^\dagger}^t[E_{c^\dagger}^t[\Pi_t(\beta_t)|Z_t\chi_t, \beta_t]|Z_t\chi_t]. \quad (63)$$

Now, β_t is independent of $Z_t\chi_t$. So, for the outer expectation to be at a maximum, its contents must be maximized for all β_t in the support of F . So, at the maximum $c^\dagger = \tilde{c}(Z_t\chi_t)$. Hence $\tilde{c} = c$. ■

A.3 Proof of Theorem 3.7

The theorem states: Consider a buy-sell symmetric equilibrium where $\pi \in (0, 1)$. The *ex ante* surplus of player t is

$$E(|\beta_t|) - \frac{k}{2} \left(\frac{\frac{k}{2} + \delta F^{-1} \left[\frac{3}{4} \right]}{\frac{k}{2} + F^{-1} \left[\frac{3}{4} \right]} \right), \quad (64)$$

where

$$\delta = \frac{E[|\beta_t| : |\beta_t| < F^{-1} \left[\frac{3}{4} \right]]}{F^{-1} \left[\frac{3}{4} \right]}. \quad (65)$$

Proof. The *ex ante* payoff to player t is $E[\Pi(\beta_t)]$. Since $\pi \in (0, 1)$, the player is indifferent to the realization of Z_t . So

$$E[\Pi(\beta_t)] = E[\Pi(\beta_t)|Z_t = 0]. \quad (66)$$

The right hand side of this is, from Theorem 3.6 and (14) – (18):

$$\frac{1}{4} E \left(\beta_t - \frac{k}{2} \mid Z_t = 0, or_t = buy \right) \quad (67)$$

$$+ \frac{1}{4} E \left(-\beta_t - \frac{k}{2} \mid Z_t = 0, or_t = sell \right) \quad (68)$$

$$+ \frac{1}{4} E \left(\left[\beta_t + \frac{k}{2} \right] e^{-\rho T_t^B} \mid Z_t = 0, or_t = bid \right) \quad (69)$$

$$+ \frac{1}{4} E \left(\left[-\beta_t + \frac{k}{2} \right] e^{-\rho T_t^A} \mid Z_t = 0, or_t = ask \right). \quad (70)$$

or, exploiting the buy-sell symmetry,

$$\frac{1}{2} E \left(\beta_t - \frac{k}{2} \mid Z_t = 0, or_t = buy \right) \quad (71)$$

$$+ \frac{1}{2} E \left(\left[\beta_t + \frac{k}{2} \right] e^{-\rho T_t^A} \mid Z_t = 0, or_t = ask \right). \quad (72)$$

Rewriting, this is

$$\frac{1}{2} E (\beta_t | Z_t = 0, or_t = buy) - \frac{k}{4} \quad (73)$$

$$+ \frac{1}{2} E (\beta_t | Z_t = 0, or_t = bid) + \frac{k}{4} \quad (74)$$

$$- \frac{1}{2} E \left(\left[\beta_t + \frac{k}{2} \right] (1 - e^{-\rho T_t^A}) | Z_t = 0, or_t = ask \right). \quad (75)$$

The first two lines of this sum to $E(|\beta_t|)$. So, the whole expression is

$$E(|\beta_t|) - \frac{1}{2} \left[\delta F^{-1} \left(\frac{3}{4} \right) + \frac{k}{2} \right] E \left[1 - e^{-\rho T_t^A} | Z_t = 0, or_t = ask \right], \quad (76)$$

and, finally, from (39),

$$E \left[1 - e^{-\rho T_t^A} | Z_t = 0, or_t = ask \right] = 1 - E_{c_{ask}}^t [e^{-\rho T_t^A}] = \frac{k}{\frac{k}{2} - F^{-1} \left[\frac{1}{4} \right]}. \quad (77)$$

■

A.4 Proof of Proposition 3.1

The proposition states: For all $\pi \in (0, 1)$, there exists an x , such that a buy-sell symmetric equilibrium exists where $\pi_t = \pi$ for all t .

Proof. Set $\beta^+ = +\infty$. Fix the initial condition $\chi_0 = (1, 0, \dots, 0, -1)'$. Consider any $\pi \in (0, 1)$.

Another equilibrium in a similar game To make applicable a theorem of Rieder's, I adapt the game. The game remains the same except where now stated.

At one point in the original game players receive the signal $Z_t \chi_t$. In the adapted game, something different happens at this point: players are informed of some z such that $Z_t \chi_t \in (z, -rev(z))$ – and they do not learn $Z_t \chi_t$ itself. Player actions are a pair,

$$(\pi_t^\#, c_t^\#), \quad (78)$$

where $c_t^\# \in \mathbb{R}^3$ is a vector of three weakly increasing elements. Also,

$$c_t = c_t^\# \quad \text{if} \quad Z_t \chi_t = z, \quad \text{and} \quad (79)$$

$$c_t = -rev(c_t^\#) \quad \text{if} \quad Z_t \chi_t = -rev(z). \quad (80)$$

So player t cannot distinguish $Z_t \chi_t$ from $-rev(Z_t \chi_t)$. In the latter case, (80) constrains her to select c_t to be the the buy-sell opposite of her selection of c_t in the former case.

Furthermore, players choose actions subject to the restrictions:

$$\pi_t^\# = \pi \quad \text{and} \quad c_t^1(0) = F^{-1} \left(\frac{1}{4} \right). \quad (81)$$

Define for player t ,

$$\hat{\beta}_t = E_{c_t}^t[\beta_t | or_t]. \quad (82)$$

Equilibrium in the adapted game involves an equilibrium plan $(\pi, c^\#(\cdot))$, such that for all $z \in [\mathbb{Z}^{(Q-M+1)}]$,

$$E_{c_t}^t[\Pi_t(\hat{\beta}_t) | Z_t \chi_t = z] \quad (83)$$

exists and is maximized when $c_t^\# = c^\#(z)$. So this replaces $\hat{\beta}_t$ with β_t in players' objective functions, and disregards the choice of π_t . No statement is made about players' choice of c_t when $Z_t = 0$.

Payoffs, $\{\Pi_t(\hat{\beta}_t) : t \in \mathbb{N}\}$, thus defined, are measurable with respect to $\mathcal{F}^{\chi, Z}$ (but are not adapted to it). Rieder (1979) proves the existence of a stationary equilibrium with Markov strategies when there are a countable number of players, a countable state space, and a compact action space. This proof applies to this equilibrium. Goettler *et al.* (2004) and (2005) previously highlighted the applicability of Rieder (1979) in dynamic trading games of this sort.

Claim : $c^\#$ gives an equilibrium in the original formulation of the game.

From now on, I only consider the original formulation of the game when $c_t = c^\#(Z_t \chi_t)$ and $\pi_t = \pi$. I show that this is an equilibrium as defined in Section 2.5. I show at the end of this proof that for all $c^\dagger$,

$$E_{c^\dagger}^t[\Pi_t(\hat{\beta}_t) | Z_t \chi_t] = E_{c^\dagger}^t[\Pi_t(\beta_t) | Z_t \chi_t]. \quad (84)$$

Furthermore, adapting the proof of Lemma 2.3 to this case, we have that for all $c^\dagger$ and for all $z \in [\mathbb{Z}^{(Q-M+1)}]$,

$$E_{-rev(c^\dagger)}^t[\Pi_t(\beta_t) | Z_t \chi_t = -rev(z)] = E_{c^\dagger}^t[\Pi_t(\beta_t) | Z_t \chi_t = z]. \quad (85)$$

Therefore, (83) and (85) imply that by varying $c^\#$, players can obtain an action c_t which is a best response in the game, regardless whether χ_t or $-rev(\chi_t)$ is true.

So $(\pi, c^\#)$ is a strategy profile which gives a buy-sell symmetric equilibrium in the original formulation of the game when actions are subject to the constraints in (81). The proof is completed by showing that these constraints are not binding.

Lemma A.1 then shows that in the original formulation of the game when $c_t = c^\#(Z_t \chi_t)$ and $\pi_t = \pi$, when actions are subject to the restrictions on a_t in (81), χ_t is stationary.

Best response when $Z_t = 0$ Since the equilibrium is stationary, the decision problem when $Z_t = 0$ is well-defined. Tracing through its proof, it can be seen that Theorem 3.6 applies to the equilibrium. So, the constraint on players' actions in (81), that $c^1(0) = F^{-1}(\frac{1}{4})$, is not binding.

The choice of π_t Since she could always choose to ignore her signal, there is a weak benefit to player t of drawing $Z_t = 1$ rather than $Z_t = 0$. Set x equal to this benefit. Then the restriction in (81), that $\pi_t = \pi$, is not binding.

I complete the current proof by showing that (84) is true. Now,

$$E_{c^\dagger}^t[\Pi_t(\beta_t)|Z_t\chi_t] = E_{c^\dagger}^t[E_{c^\dagger}^t[\Pi_t(\beta_t)|Z_t\chi_t, or_t]|Z_t\chi_t], \quad (86)$$

and, conditional on or_t , $\Pi_t(b)$ is linear in b , so this is

$$E_{c^\dagger}^t[\Pi_t(\beta_t)|Z_t\chi_t] = E_{c^\dagger}^t[E_{c^\dagger}^t[\Pi_t(E_{c^\dagger}^t[\beta_t|Z_t\chi_t, or_t])|Z_t\chi_t, or_t]|Z_t\chi_t] \quad (87)$$

$$= E_{c^\dagger}^t[E_{c^\dagger}^t[\Pi_t(E_{c^\dagger}^t[\beta_t|or_t])|Z_t\chi_t, or_t]|Z_t\chi_t] \quad (88)$$

$$= E_{c^\dagger}^t[E_{c^\dagger}^t[\Pi_t(\hat{\beta}_t)|Z_t\chi_t, or_t]|Z_t\chi_t] \quad (89)$$

$$= E_{c^\dagger}^t[\Pi_t(\hat{\beta}_t)|Z_t\chi_t], \quad (90)$$

where (88) follows because β_t and $Z_t\chi_t$ are independent random variables. ■

A.5 Stationarity Lemma

Lemma A.1 *In the proof of Proposition 3.1, χ_t is stationary in the buy-sell symmetric equilibrium when actions are subject to the constraints in (81). The equilibrium is the one given by strategy profile $c_t = c^\#(Z_t\chi_t)$ and $\pi_t = \pi$.*

Proof.

The object of interest is a buy-sell symmetric equilibrium in the original game, where actions are subject to constraints: the constraints are that

$$\pi_t = \pi \quad \text{and} \quad c(0) = \begin{pmatrix} F^{-1}(\frac{1}{4}) \\ 0 \\ F^{-1}(\frac{3}{4}) \end{pmatrix}.$$

I make use of Proposition 3.2, which, tracing through its proof, is true for this case, but applies only when $Z_t = 1$. I use the following condition for stationarity without proof:

Condition A.2 Let W_t , defined for $t \in \mathbb{N}$, be a Markov process with state space contained in $\mathbb{Z}$, where for all W_t the conditional support of $(W_{t+1} - W_t)$ is $\{-1, 0, 1\}$. Suppose that there exists a threshold $L^* \in \mathbb{N}$ and $\eta \in \mathbb{R}^+$ such that

$$E[|W_{t+1}| \mid W_t] \leq |W_t| - \eta$$

whenever $|W_t| > L^*$. Then a distribution may be given to W_0 so that W_t is stationary.

The intuition for this condition is that the process reverts toward zero outside $[-L^*, L^*]$, so that L^* and $-L^*$ are positive recurrent. It is now applied in various cases when W_t is a function of the offered depth at time t .

A.5.1 First, let $W_t = \min(\chi_0, 0)$.

So W_t is non-zero only in the artificial case of a negative depth at B_t . Then 1) no player, knowing that the bid depth is negative, would submit a *buy* and 2) if $\beta_t = 0$ player t would submit a *bid*. So

$$\Pr[or_t = bid \mid Z_t = 1] > \frac{1}{2}, \quad (91)$$

while only types below $-\frac{k}{2}$ can *sell*, so

$$\Pr[or_t = sell \mid Z_t = 1] \leq F\left(-\frac{k}{2}\right). \quad (92)$$

Players with $Z_t = 0$ submit *bids* and *sells* with equal probability. Hence whenever $\chi_t < 0$, recalling the definition of g at the endpoints Mk and Qk ,

$$E[\min(\chi_{t+1}, 0) \mid \min(\chi_t, 0)] > \min(\chi_t, 0) + \pi \left(\frac{1}{2} - F\left(-\frac{k}{2}\right) \right)$$

As $\pi > 0$ the negative depth has a conditionally upward drift. A similar argument applies when there is negative depth at A_t . ■

A.5.2 Second, let $W_t = \min(L_t^{A_t+}, L_t^{B_t+})$.

Recall that $L_t^{A_t+}$ is $\max\{L_t^{A_t} + 1, 0\}$, and $L_t^{B_t+}$ is $\max\{L_t^{B_t} + 1, 0\}$.

For any natural number L , let $LA(L, t) \subset \Omega$ be the set

$$LA(L, t) = \{\omega \in \Omega : W_t = L, or_t = ask\}. \quad (93)$$

consider the quantity

$$\sup\{E[e^{-\rho T_t^A} \mid \chi_t, or_t](\omega) : \omega \in LA(L, t)\}. \quad (94)$$

For all t this quantity converges to zero as $L \rightarrow \infty$ since time to execution is at most as fast as if all future traders submitted *buys*, a time which becomes arbitrarily distant as $L \rightarrow \infty$. Hence from (26),

$$\inf\{c^1(\chi_t) : W_t = L\} \quad (95)$$

converges to $-\frac{k}{2}$ (from below) as $L \rightarrow \infty$, and so

$$\inf\{F(c^1(\chi_t)) : W_t = L\} \quad (96)$$

converges to $F(-\frac{k}{2})$, which is $> \frac{1}{3}$. Hence there exist L^* and $\alpha > \frac{1}{3}$ such that for all $L > L^*$,

$$\Pr[or_t = sell | Z_t = 1, W_t = L] > \alpha > \frac{1}{3}. \quad (97)$$

Considering the equivalent argument on the other side of the book, and (for the latter two inequalities) that *bid*, *ask*, *sell* and *buy* are mutually exclusive, there also exist L^* and $\eta > 0$ such that for all $L > L^*$,

$$\Pr[or_t = sell | Z_t = 1, W_t = L] > \frac{1}{3} + \eta, \quad (98)$$

$$\Pr[or_t = buy | Z_t = 1, W_t = L] > \frac{1}{3} + \eta, \quad (99)$$

$$\Pr[or_t = bid | Z_t = 1, W_t = L] < \frac{1}{3} - \eta, \quad (100)$$

$$\Pr[or_t = ask | Z_t = 1, W_t = L] < \frac{1}{3} - \eta. \quad (101)$$

Therefore there exists an $\eta^* > 0$ such that if $W_t > L^*$,

$$E[W_{t+1} | W_t, Z_t = 1] < W_t - \eta^*. \quad (102)$$

But as, when $Z_t = 0$, *bid*, *ask*, *sell* and *buy* are played with equal probability,

$$E[W_{t+1} | W_t, Z_t = 0] = W_t. \quad (103)$$

Combining the last two equations using the Law of Iterated Expectations,

$$E[W_{t+1} | W_t, Z_t = 1] < W_t - \pi\eta^*. \quad \blacksquare \quad (104)$$

A.5.3 Third, let $W_t = |L_t^{A_t+} - L_t^{B_t+}|$

So, disregarding negative depths, W_t is the absolute difference between the depth at the best bid, and the the depth at the best ask. As the equilibrium is buy-sell symmetric, consider w.l.o.g. the case where $L_t^{A_t+} > L_t^{B_t+}$. For any natural number L , let $LA(L, t) \subset \Omega$ be the set

$$LA(L, t) = \{\omega \in \Omega : W_t = L, or_t = ask\}. \quad (105)$$

consider the quantity

$$\sup\{E[e^{-\rho T_t^A}|\chi_t, or_t](\omega) : \omega \in LA(L, t)\}. \quad (106)$$

For all t this quantity converges to zero as $L \rightarrow \infty$ since time to execution is at most as fast as if all future traders submitted *buys*, a time which becomes arbitrarily distant as $W_t \rightarrow \infty$. Hence from (26),

$$\inf\{c^1(\chi_t) : W_t = L\} \quad (107)$$

converges to $-\frac{k}{2}$ (from below) as $L \rightarrow \infty$, and so

$$\inf\{F(c^1(\chi_t)) : W_t = L\} \quad (108)$$

converges to $F(-\frac{k}{2})$ from below. Let $c^2(\chi_t)$ be the second element of $c(\chi_t)$. Then,

$$\left(\frac{k}{2} - c^2(\chi_t)\right) E_{c_{ask}}^t[e^{-\rho T_t^A}|\chi_t] = \left(\frac{k}{2} + c^2(\chi_t)\right) E_{c_{bid}}^t[e^{-\rho T_t^B}|\chi_t]. \quad (109)$$

Now, $E_{c_{bid}}^t[e^{-\rho T_t^B}|\chi_t]$ is at most as great as $e^{-\rho \Delta L_t^{Bt+}}$ since time to execution is at most as fast as if all future traders submitted *sells*. And, $E_{c_{ask}}^t[e^{-\rho \Delta T_t^A}|\chi_t]$ is larger than $e^{-\rho L_t^{At+} F(-\beta^+)}$, since for all χ_t there is probability of at least $F(-\beta^+)$ of a trader arriving who submits a *buy*. Using these inequalities,

$$\left(\frac{k}{2} - c^2(\chi_t)\right) e^{-\rho \Delta L_t^{At+} F(-\beta^+)} > \left(\frac{k}{2} + c^2(\chi_t)\right) e^{-\rho \Delta L_t^{Bt+}} \quad (110)$$

so

$$e^{-\rho \Delta (L_t^{At+} F(-\beta^+) - L_t^{Bt+})} > 1 + \frac{2c^2(\chi_t)}{\left(\frac{k}{2} - c^2(\chi_t)\right)}. \quad (111)$$

Under the maintained assumption that $L_t^{At+} > L_t^{Bt+}$,

$$W_t F(-\beta^+) < L_t^{At+} F(-\beta^+) - L_t^{Bt+}. \quad (112)$$

So,

$$\sup\left\{\frac{2c^2(\chi_t)}{\left(\frac{k}{2} - c^2(\chi_t)\right)} : W_t = L\right\} < e^{-\rho \Delta L F(-\beta^+)} - 1, \quad (113)$$

and, via an increasing transformation,

$$\sup\{F(c^2(\chi_t)) : W_t = L\} < F\left(\frac{\frac{k}{2} \left(e^{-\rho \Delta L F(-\beta^+)} - 1\right)}{2 + e^{-\rho \Delta L F(-\beta^+)} - 1}\right), \quad (114)$$

so that

$$\sup\{F(c^2(\chi_t)) : W_t = L\} \rightarrow F\left(-\frac{k}{2}\right), \quad (115)$$

as $L \rightarrow \infty$. Combining (115) and (108),

$$\lim_{L \rightarrow \infty} \Pr[or_t = ask | Z_t = 1, W_t = L] = 0. \quad (116)$$

But for all χ_t there is probability of at least $F(-\beta^+)$ of a trader arriving who submits a *buy*.

Therefore there exists an $\eta^* > 0$ such that if $W_t > L^*$,

$$E[W_{t+1} | W_t] < W_t - \eta^*. \quad \blacksquare \quad (117)$$

A.5.4 Conclusion to the proof of the Stationarity Lemma

The first two of the three previous parts showed that $\min(L_t^{B_t}, 0)$ and $\min(L_t^{A_t}, 0)$ (can be given appropriate distributions on their initial conditions so that they) are stationary; and that $\min(L_t^{A_t+}, L_t^{B_t+})$ is stationary. Thus $\min(L_t^{B_t}, L_t^{A_t})$ is stationary. The third part showed that $|L_t^{A_t+} - L_t^{B_t+}|$ is stationary. Combining these results, we have that $L_t^{B_t}$ and $L_t^{A_t}$ are stationary, and therefore have a positive mass in their stationary distributions at zero. Therefore A_t and B_t have a positive mass in their stationary distributions on all values in the admissible range $[Mk, Qk]$. Therefore, χ_t is also stationary. $\blacksquare$